A randomized-and-derandomized subprocedure that finds a large vertex set meeting every edge of a 2-3-uniform hypergraph in the required parity

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Allowed standard prerequisites (editorial).

- The linearly ordered (LO) colouring of a hypergraph and the class of LO k-colourable hypergraphs. Defined in the introduction of the paper, before source line 100; the counted statement uses the case k = 2. Fixes the notion of an LO 2-colouring that the input hypergraph is assumed to admit.
- The linear system Av = 1^m attached to an LO 2-colourable hypergraph, and the statement that it does not fix the value of any variable. Set up in the section introducing the algorithm based on equations modulo 2 (source lines 147-150); the hypothesis of the counted statement is exactly

this non-fixing condition. Supplies the affine solution space in which the randomised choice of a_i is made, and the fact that for every vertex x some homogeneous solution v^i has v^i_x = 1.

Lemma 1. *There is an algorithm which, if given a $\{2,3\}$-uniform hypergraph H with n vertices and m edges that admits an LO 2-colouring and such that the implied linear system of equations $Av = 1^m$ does not fix the value of any variable, outputs a subset T of vertices that intersects edges of size three in zero or two vertices and edges of size two in exactly one vertex. Moreover, we have $|T| \geq n/2$. The algorithm runs in $O(n^3 + nm)$ time.*

Proof. We first describe a randomised version of our algorithm, and then derandomise it. The set of solutions to $Av = 1^m$ is an affine space and hence a generic solution can be written as $v = v^0 + \sum_{i=1}^{r} a_i v^i$ for a basic solution v^0, linearly independent solutions to the homogeneous system v^i, and field elements (in this case bits) a_i. The fact that no variable is fixed implies that for each vertex x there is some positive i such that $v^i_x = 1$.

For the randomised algorithm choose $a_1, \ldots, a_r$ to be independent identically distributed uniformly random bits, and set T to be the set of variables x, such that $v_x = 0$. Clearly T satisfies the conclusion of the lemma as in each edge we have an odd number of ones. For every vertex x, there exists positive i such that $v^i_x = 1$ — hence, due to the influence of $a_i v^i_x$, we see that x is included in T with probability $\frac{1}{2}$. Thus, on average T contains half the vertices.

Now, we derandomise this algorithm using the method of conditional expectations. Go through the variables a_i in increasing order and fix its value once and for all. Fixing the value of a_i determines the value(s) of some v_x while other values remain undetermined. For each value being determined $v^i_x = 1$ and hence one value of a_i gives the final value 0 and the other gives final value 1. Set a_i such that at least half the determined values are 0. After we have fixed all a_i this way, we have a final solution with at least $n/2$ zeroes.

The bottleneck of the running time of this algorithm is solving the linear system of equations. This can be done in the advertised running time since every equation has $O(1)$ entries. □